

ADVANCED SUBNETTING AND VLSM

1. Given a company with 3 departments (HR: 20 users, Sales: 50 users, Tech: 100 users), assign appropriate subnets using VLSM from the 192.168.10.0/24 network and write the subnet address, subnet mask, and usable IP range for each department.

- -> Network: **192.168.10.0/24**
- HR: 20 users
- Sales: 50 users
- Tech: 100 users

VLSM Calculation

Department	Hosts Needed	CIDR	Subnet Mask	Subnet Address	Usable IP Range
Tech	100	/25	255.255.255.128	192.168.10.0	192.168.10.1 – 192.168.10.126
Sales	50	/26	255.255.255.192	192.168.10.128	192.168.10.129 – 192.168.10.190
HR	20	/27	255.255.255.224	192.168.10.192	192.168.10.193 – 192.168.10.222

2. You are given three office locations (Ahmedabad, Surat, Rajkot). Each office has three departments: Admin (10 users), Marketing (30 users), and Support (60 users). Design an IP addressing scheme using VLSM for all offices using the 10.0.0.0/22 block, ensuring no overlap and efficient usage. List each subnet with its office, department, subnet mask, and first/last usable IP.

->Base network: **10.0.0.0/22**

The /22 provides addresses from:

10.0.0.0 – 10.0.3.255

For each office:

- Support: 60 users → **/26** → 62 usable
- Marketing: 30 users → **/27** → 30 usable
- Admin: 10 users → **/28** → 14 usable

A convenient design is to give each office its own /24-sized section, while using VLSM inside it.

Ahmedabad — 10.0.0.0/24

Department	Subnet	Mask	First Usable	Last Usable
Support	10.0.0.0/26	255.255.255.192	10.0.0.1	10.0.0.62
Marketing	10.0.0.64/27	255.255.255.224	10.0.0.65	10.0.0.94
Admin	10.0.0.96/28	255.255.255.240	10.0.0.97	10.0.0.110

Surat — 10.0.1.0/24

Department	Subnet	Mask	First Usable	Last Usable
Support	10.0.1.0/26	255.255.255.192	10.0.1.1	10.0.1.62
Marketing	10.0.1.64/27	255.255.255.224	10.0.1.65	10.0.1.94
Admin	10.0.1.96/28	255.255.255.240	10.0.1.97	10.0.1.110

Rajkot — 10.0.2.0/24

Department	Subnet	Mask	First Usable	Last Usable
Support	10.0.2.0/26	255.255.255.192	10.0.2.1	10.0.2.62
Marketing	10.0.2.64/27	255.255.255.224	10.0.2.65	10.0.2.94
Admin	10.0.2.96/28	255.255.255.240	10.0.2.97	10.0.2.110

This leaves **10.0.3.0/24** completely available for future offices or expansion.

3. Route Summarization

Given:

- Mumbai → **172.16.10.0/24**
- Pune → **172.16.11.0/24**
- Baroda → **172.16.12.0/24**

The smallest single supernet containing all three is:

172.16.8.0/21

A /21 covers:

172.16.8.0 – 172.16.15.255

Therefore, all three office networks fall inside:

Summarized route: 172.16.8.0/21

**3. A startup has offices in Mumbai, Pune, and Baroda. Each office will connect to HQ via VPN. Propose a route summarization for all three offices if their assigned subnets are: Mumbai (172.16.10.0/24), Pune (172.16.11.0/24), Baroda (172.16.12.0/24). Write the summarized route and explain why it is efficient.

Hint: Find the smallest supernet that includes all three subnets.**

->Given:

- Mumbai → **172.16.10.0/24**
- Pune → **172.16.11.0/24**
- Baroda → **172.16.12.0/24**

The smallest single supernet containing all three is:

172.16.8.0/21

A /21 covers:

172.16.8.0 – 172.16.15.255

4. Given a scenario where the Tech department in Ahmedabad grows from 100 to 200 users, update your previous VLSM plan to accommodate the new size with minimal changes to the rest of the network. List the new subnet details and

explain your approach.

Constraint: Do not change the existing subnets for HR or Sales unless absolutely necessary.

->Original plan:

- Tech: 100 users → 192.168.10.0/25
- Sales: 192.168.10.128/26
- HR: 192.168.10.192/27

A 200-user department requires at least 200 usable addresses.

A /24 provides:

256 total – 2 = 254 usable addresses

So Tech needs:

192.168.11.0/24

However, there is an important limitation: 192.168.10.0/24 cannot contain a /24 Tech subnet while also retaining the existing HR and Sales subnets.

Therefore, to satisfy the constraint of keeping HR and Sales unchanged, the practical solution is to expand the available address space to 192.168.10.0/23.

Updated plan

Departme nt	User s	Network	Mask	Usable Range
Sales	50	192.168.10.128/ 26	255.255.255.19 2	192.168.10.12 9 – 192.168.10.19 0
HR	20	192.168.10.192/ 27	255.255.255.22 4	192.168.10.19 3 – 192.168.10.22 2
Tech	200	192.168.11.0/24	255.255.255.0	192.168.11.1 – 192.168.11.25 4

5. Create 5 VLSM subnetting challenge questions inspired by real apps you use (e.g., Zomato delivery zones, Flipkart warehouse locations, or IPL team fan clubs). For each, provide the required number of hosts and a base network address.

-> **Challenge 1 — Zomato Delivery Zones** 🍔

A Zomato regional office has three delivery zones:

- Central Zone: **100 devices**
- North Zone: **50 devices**
- South Zone: **20 devices**

Base network: **192.168.50.0/24**

Challenge 2 — Flipkart Warehouse 🛒

A Flipkart warehouse has:

- Packing: **80 hosts**
- Inventory: **40 hosts**
- Administration: **15 hosts**
- Security: **10 hosts**

Base network: **10.20.0.0/23**

Challenge 3 — IPL Fan Club 🏏

An IPL fan-club organization has separate networks for:

- Mumbai Fans: **120 hosts**
- Chennai Fans: **60 hosts**
- Bengaluru Fans: **30 hosts**
- Hyderabad Fans: **10 hosts**

Base network: **172.20.0.0/23**

Challenge 4 — Swiggy Restaurant Network 🍕

A large Swiggy partner restaurant needs networks for:

- POS terminals: **25 devices**

- Kitchen systems: **15 devices**
- Staff Wi-Fi: **50 devices**
- Guest Wi-Fi: **100 devices**

Base network: **192.168.100.0/24**

Challenge 5 — Amazon Delivery Hub

An Amazon delivery hub requires:

- Delivery scanners: **70 devices**
- Warehouse computers: **45 devices**
- Manager/Admin systems: **12 devices**
- CCTV network: **30 devices**

Base network: **10.50.0.0/23**